Late cosmic acceleration as a reading of the observer's clock tied to the growth of structure: registered tests against DESI DR2, supernovae, CMB distance priors and growth data

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

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Abstract

In its own combinations with the CMB and supernovae, DESI DR2 prefers an evolving dark energy with w0 > −1 and wa < 0 over ΛCDM at 2.8–4.2σ (DESI Collaboration 2025). This record tests the dark-energy history of the author's framework, Mathematical Materiality (MM), against DESI DR2 BAO, compressed CMB distance priors (Chen, Huang and Wang 2019) and three supernova samples, and against the growth of structure. It then proposes a phenomenological hypothesis, read with MM by coherence: the late acceleration as a reading of the observer's clock that changes with the growth of cosmic structure. Every test was registered with SHA-256 fingerprints before its computation (knowing earlier results on the same data, as each registration states); additions decided after seeing results are labelled; failed predictions are reported. An independent AI system checked each block in one to nine rounds. It is an exploratory, preliminary study.

Version 2.50 of the Mathematical Materiality (MM) series, 14 October 2026. Continues MM 2.32 (*From Relational Links to Dark Energy*, DOI 10.5281/zenodo.22839525). Author: Ugo Lanciano (independent researcher, not a professional physicist). Computations and texts were prepared with AI assistants (Claude; gpt-6-astra via Codex as an independent checker); they have not been reviewed by a human referee. Original text and materials: CC BY 4.0. Original code: MIT licence.

At a glance

The question. In the combinations analysed by DESI (2025), with the CMB and supernovae, the CPL model of evolving dark energy, w(a) = w0 + wa (1 − a), is preferred over ΛCDM, the standard model with a cosmological constant (w = −1). Can this late acceleration be read as an effect of how an observer living inside growing structures reads time, rather than as an extra substance?

The idea in three lines. Take a universe of matter and radiation only, with no dark energy. Let the expansion rate *read by the observer* be higher than the bare one by a factor 1 + δ0 S(z). S(z) is a global tracer of the growth of cosmic structure, normalized to 1 today and tending to 0 in the limit without structure; mostly S = [D(z)/D(0)]^n, where D is the linear growth factor. The clock model is a different expansion history from the published MM 2.32; a translation of the 2.32 record counts into this clock law has not yet been constructed.

Main results. Data: DESI DR2 baryon acoustic oscillations (BAO), compressed cosmic microwave background (CMB) distance priors and one supernova sample at a time (Pantheon+ / Union3 / DES-Dovekie). Δχ² is relative to ΛCDM; negative is better. σ8 is the amplitude of matter fluctuations today on 8 h^{-1} Mpc.

what	result
MM 2.32 tested through three registered readings, all with w ≥ −1	"in tension" overall, 2.1–3.1σ (details in section 4)
parametric reconstructions without MM, with ΛCDM-type early physics	in the parametrizations tried, the best-localized deviations are at z ≈ 0.4–1; no significant extra factor is found on the CMB distance
the clock idea, with S = [D/D0]^n and the ΛCDM growth curve D	Δχ² = −6.6 / −11.6 / −10.3; CPL has χ² lower by 0.6–2.3, with one more parameter
the clock follows the growth its own history	Δχ² = −7.0 / −11.9 / −10.6; CPL has χ² lower by 0.3–2.0

produces (no external growth curve)	
growth predicted by that self-consistent version, with Planck's amplitude	σ8 = 0.797–0.808, compatible with SDSS, DESI and 6dFGS growth; χ^2 close to ΛCDM's
effective equation of state of the clock histories	w ≈ −0.8 today, crossing −1 between z ≈ 0 and 0.6, the shape favoured by DESI's w0wa fits
growth of structure for the MM 2.32 family	with Planck's primordial amplitude fixed, compatible in most tested cases; not distinguished from ΛCDM by these measurements; median σ8 = 0.790 along the chain (exceptions and interval: section 8)

What is fitted or assumed.

- The law linking growth and clock is assumed: the model is phenomenological, not a derivation from MM principles.
- The exponent n ≈ 3.6–3.7 is fitted, not derived. The amplitude δ0 is fixed by today's normalization, not predicted.
- The rule that distances follow the read rate is an extra assumption: a change of clock alone would not change measured distances.
- Growth and expansion data are partly correlated.
- With the same cosmological ingredients and perturbation dynamics, these data cannot tell the clock reading from a substance with the same expansion history.
- The study is exploratory and preliminary.

Why it is offered. One rule answers several requirements at once: acceleration without Λ, a late onset, the evolving-w shape, and growth compatible with the data. That is coherence, not proof (section 10).

Where to look. The hypothesis is in section 7, the test of MM 2.32 in section 4, the data-only reconstruction in section 6, and the growth of the 2.32 family in section 8. Each block has a final note (NOTA_ESITO_*.md, in Italian) with every number and limit.

Labels used in this deposit

label	meaning
T1	test of the published MM 2.32 expansion history against DESI DR2 (section 4)
L1, L2, L3	the three registered readings of 2.32 in T1: CPL restricted to w ≥ −1 at all times; the registered thawing edge of CPL, w0 + wa = −1 with w0 ≥ −1, that is w(a) = −1 + (1 + w0) a; the 2.32 model itself within its parameter ranges
T2–T7	further translations of the author's working premises into expansion histories (section 5)
R1	reconstruction of what the data ask, with no MM input (section 6)
T8, T8a–T8g	the steps of the clock hypothesis (section 7): T8a what the data ask without Λ; T8 three tracers; T8c growth; T8d–T8e tracer from measured growth; T8f check with computed growth; T8g self-consistent version
K1, K2, K3	tracers S(z): K1 collapsed mass fraction; K2 ΛCDM linear growth to a power n; K3 growth of a universe without dark energy (S = a^n)
G0, B1–B3	growth of structure for the 2.32 family: G0 for selected histories from T1–T7; B1 check of the coupled MM Boltzmann code; B2 noise of the readings; B3 growth along the 2.32 chain (section 8)
C1, C2, C3	data combinations: C1 = DESI BAO + CMB priors; C2 = supernovae + CMB priors; C3 = all three (the main one)
PP, U3, DV, D5	supernova samples: Pantheon+, Union3, DES-Dovekie; DES-SN5YR as a check only
CHW	compressed CMB distance priors of Chen, Huang and Wang (2019)
S1, S2, S3	growth data sets: S1 SDSS; S2 DESI DR1 full-shape and peculiar velocities; S3 6dFGS

holds / in tension / disfavoured / excluded	registered decision zones, usually $\Delta\chi^2$ vs ΛCDM below 4 / 4–9 / 9–25 / above 25; labels of a criterion, not probabilities. T1 uses its own simulation-based rule (section 4)
fast code / full check	the fast code fits; the full check re-evaluates the fitted points with CAMB 2.0.4 and HyRec-2

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1. What this deposit is, and what it is not

It is **one link** in the MM programme: dark energy. It is **not** a derivation of dark energy from first principles, and it does **not** show that the clock reading is physically distinct from a dark-energy component with the same expansion history: with the ingredients and perturbation rule adopted here, the two give the same growth (section 7.4).

How the work was done.

- **Registered first:** each test was registered before its computation (translation, data, decision zones and the author's predictions), with a SHA-256 fingerprint and a UTC time in a REGISTRO.txt. Registrations were written knowing the results of earlier tests on the same data; each says what was known.
- **Additions and failures:** additions decided after seeing results are labelled as such; failed predictions are reported as failed.
- **Independent check:** every block was checked by gpt-6-astra via Codex, in one to nine rounds. Its reports and the corrections are in the records (section 12).

2. Prior work, first

topic	published result used here
BAO and evolving dark energy	DESI Collaboration, DESI DR2 Results II, arXiv:2503.14738 (2025): in its own combinations with the CMB and supernovae, w0waCDM is preferred over ΛCDM at 2.8–4.2σ, with $w_0 > -1$ and $w_a < 0$
growth from full-shape clustering	DESI Collaboration, DESI 2024 V, arXiv:2411.12021; DESI peculiar velocities, arXiv:2512.03230
SDSS BAO and growth	eBOSS Collaboration, Alam et al., PRD 103, 083533 (2021); BOSS DR12, Alam et al., MNRAS 470, 2617 (2017); 6dFGS, Beutler et al., arXiv:1204.4725
supernovae	Pantheon+, Brout et al., ApJ 938, 110 (2022); Union3, Rubin et al., arXiv:2311.12098; DES-Dovekie, Popovic et al., arXiv:2511.07517, which reports a reduced preference for evolving dark energy compared with DES-SN5YR
CMB distance priors	Chen, Huang, Wang, JCAP 02, 028 (2019), arXiv:1808.05724, derived in ΛCDM from Planck 2018 TT,TE,EE+lowE without the lensing-reconstruction likelihood; Planck 2018 VI, arXiv:1807.06209
weak lensing S8	KiDS-Legacy, arXiv:2503.19441; DES Y3, arXiv:2105.13549; HSC Y3, arXiv:2304.00701
parametrizations	Chevallier, Polarski, IJMPD 10, 213 (2001); Linder, PRL 90, 091301 (2003) (CPL); thawing and freezing fields, Caldwell, Linder, PRL 95, 141301 (2005)
crossing $w = -1$	Vikman, PRD 71, 023515 (2005); Hu, PRD 71, 047301 (2005); PPF treatment, Fang, Hu, Lewis, PRD 78, 087303 (2008)
clocks of observers in structures	timescape cosmology, Wiltshire, NJP 9, 377 (2007) and arXiv:0909.0749 (2009); Seifert et al., MNRAS Letters 537, L55 (2025), reporting with Pantheon+ $\ln B > 5$ for timescape over ΛCDM on the full sample and $\ln B > 1$ with $z > 0.075$ (a result of that comparison, not of the model here)
backreaction	Buchert, Gen. Rel. Grav. 32, 105 (2000), arXiv:gr-qc/9906015: averaged equations of an inhomogeneous universe contain terms absent in Friedmann's
collapsed fraction, transfer function	Press, Schechter, ApJ 187, 425 (1974); Eisenstein, Hu, ApJ 496, 605 (1998)
codes	CAMB, Lewis, Challinor, Lasenby, ApJ 538, 473 (2000) (version 2.0.4); HyRec-2, Lee, Ali-Haïmoud, PRD 102, 083517

The closest precedent is timescape. It attributes apparent acceleration to the difference between the clocks of observers in bound structures and the volume-average clock, with walls, voids and Buchert averaging. The model here uses neither voids nor averaging: it is a one-function relation between the observer's read expansion rate and a tracer of structure growth.

3. Data and procedure

- **Data:** the combinations C1–C3 and the samples are listed in the label table. Third-party data are not redistributed; origins and SHA-256 fingerprints are in FONTI_DATI.md.
- **Codes:** a fast code (closed-form background, tabulated recombination) for the fits; a full check with CAMB 2.0.4 and HyRec-2, with the dark sector passed as a w(a) table through PPF, for the minima.
- The full check fails for K1, T8d and two variants of R1 (declared in each note).
- T8e and T8g pass it only with the density tolerance widened to 10^{-4}.
- **Approximations to keep in mind:**
- the CHW priors are derived in ΛCDM, so their use for other histories is an approximation;
- the three supernova samples share the BAO and CMB data, so they are not three independent confirmations;
- growth data are never summed with each other.

4. MM 2.32 against DESI DR2 (T1; blocchi/01_T1_tensione_DESI/NOTA_ESITO_T1.md)

- **What was tested:** the published 2.32 history has $w \geq -1$ at every epoch by construction. Three readings were registered (L1, L2, L3 in the label table).
- **How:** for L1 and L2, an analytic test statistic checked against simulations; for L3, a calibration with simulations. In total: 22 runs of 400 simulations, four extended to 2000, and 3728 tail and control simulations refitted separately. Intervals are Clopper-Pearson.
- **Result (C3):**

reading	Pantheon+	Union3	Dovekie	overall
L1	2.15σ, in tension	3.14σ, disfavoured	2.55σ, in tension	**in tension**
L2	2.08σ	3.10σ	2.50σ	**in tension**
L3 (first 400 simulations)	2.38σ	2.81σ	3.02σ	**in tension**

- None meets the registered criterion "falls with today's data".
- The L3 zones for Union3 and Dovekie are uncertain because of the finite number of simulations, and the MM calibration uses one null point.
- Without DESI BAO (C2), L1 and L2 hold; L3 is not calibrated there.
- **A correction to the 2.32 minimum:** with more starting points, $\chi^2 = 1419.66$ instead of 1420.02, at the edge of the allowed parameter range ($u_c = -3$, $\delta u = 0.1$). The improvement over ΛCDM is 1.52, not 1.17. This is not a measured preference for a late, narrow transition: widening the range, the minimum follows the new edge.

5. Further readings of the author's premises (T2–T7)

Each is a registered translation of a working premise into an expansion history. A negative result concerns that translation, not the premise. Final notes: blocchi/02_… to blocchi/06_….

test	translation	result
T2	dark energy as a distant effect of the splitting of the relational network (four laws D1–D4)	D3 equals ΛCDM by construction; the laws that add an expansion rate are disfavoured or excluded. An effect on supernova fluxes only, on top of ΛCDM (D4), recovers about a third of the w0wa improvement; without dark energy (D4b) it fits very badly
T3	the value of the dark-energy density from twelve "rate × epoch" combinations	none gives $\Omega_\Lambda \approx 0.698$; a relation with $3\sqrt{3}$ returns Ω_Λ within +0.0014/+0.0022, using flatness and the fitted parameters, but does not fix the absolute density
T4	the rate from when objects come into being and from the distance assigned to them in the tested	none of 18 combinations within 0.02 of $\Omega_\Lambda = 0.698$ (central values; not a full statistical

	translation ("diffraction distance"; see the T4 final note)	test)
T5	dark energy as a fraction of the network passed on locally, switched on at redshift z_v (three shapes)	a late switch-on of Λ holds from z_v = 1.5 on the registered grid (best $\Delta\chi^2 = -3.9$ at z_v = 2.5, still above CPL by +3.7/+10.0/+6.9). Shapes tied to the collapsed fraction or to growth as a density are excluded, except one collapsed-fraction case, which is disfavoured. The collapsed-fraction shape has no full check
T6, T7	a stretching of read distances, largest on the CMB	$\Delta\chi^2 = -4.6$ to -5.1, a part of the CPL gain. The author later withdrew the premise: section 6 puts the deviation at z ≈ 0.5–1, not on the CMB

6. What the data ask, without MM (R1; NOTA_ESITO_R1.md)

Reconstructions in redshift bands, with ΛCDM-type early physics and no MM input:

reconstruction (extra parameters)	$\Delta\chi^2$ vs ΛCDM (PP / U3 / DV)	vs CPL
one distance factor D(1 + ε(z)), nodes 0.5, 1, 2 and the CMB (4)	−9.1 / −9.2 / −9.3	−1.5 / +4.6 / +1.5
radial and transverse distances separate (7)	−13.1 / −14.1 / −13.3	−5.5 / −0.3 / −2.5
bounded dark-energy density in bands (6; fast code only)	−13.8 / −15.5 / −12.7	−6.2 / −1.6 / −1.9

- **Where:** in the parametrizations tried, the best-localized deviations are at z ≈ 0.4–1. There is no significant extra factor on the CMB distance ($\sqrt{\Delta\chi^2}$ 0.2–0.9). The dark-energy density is above today's at z ≈ 0.4 (+6/+14 %) and z ≈ 0.8 (+16/+26 %), with $\sqrt{\Delta\chi^2}$ 1.3–2.2 at the null value (local indicators, not global significances). Beyond z ≈ 1.2 it is undetermined.
- **Radial and transverse distances:** the radial one asks −3.6/−4.5 % at z ≈ 0.5 and the transverse −1.3/−2.1 % at z ≈ 1. Their difference is below 1.8 times its error at every node. The tested reconstruction with bounded dark-energy density in bands, evaluated only with the fast code, does as well with one parameter fewer. These data do not require separate radial and transverse readings.

7. The hypothesis: the read expansion rate follows the growth of structure (T8; NOTA_ESITO_T8.md and NOTA_ESITO_T8g.md)

7.1 Translation (one among several possible)

- **Bare universe:** matter and radiation (with neutrinos) and no dark energy, $H_{bare}(z) = 100\sqrt{G(z)}$ km/s/Mpc, where G is the physical density in ω units.
- **Read rate:** $H_{read}(z) = H_{bare}(z)\,[1 + \delta 0\, S(z)]$, with S(0) = 1 and $S \to 0$ without structure. δ0 is fixed by flatness, $(1 + \delta 0)^2 = h^2/G(0)$: a normalization, not a prediction of the amplitude.
- **Distances:** $D_M = \int c\, dz / H_{read}$. **This is an additional physical assumption**, so the model is phenomenological, not a derivation from clock premises.
- **Sign:** with this form the deceleration parameter is $q = q_{bare} - \delta 0\, S'/(1 + \delta 0\, S)$, where $S' = dS/d\ln a \geq 0$, so growing structure can produce acceleration. For δ0 > 0, the opposite form, $H_{bare}/(1 + \delta 0\, S)$, decelerates more than the bare universe; with the same δ0 it does not preserve today's normalization, and it is used here only to check the sign. The author's premise "proper time slows where structure grows" is read accordingly: the slowed proper time makes measured rates appear faster.
- **Equivalence:** the same H(z) can be written as an effective dark density $G[(1 + \delta 0\, S)^2 - 1]$. Standard cosmology would read it as a substance.

7.2 What the data ask without Λ (T8a)

From the ΛCDM, CPL and band-reconstruction minima: $H_{bare}(0)$ = 37.5–37.8 km/s/Mpc. The read rate must exceed the bare one by 75–82 % today, 30–32 % at z = 0.5, 12–14 % at z = 1, 3–4 % at z = 2, and about 0 by z = 5. So S(z) is 0.36–0.42 at z = 0.5 and 0.15–0.17 at z = 1.

7.3 Three tracers of structure (C3)

tracer	registered grid: $\Delta\chi^2$ vs ΛCDM	zone	best after declared refinements	vs CPL

	(PP / U3 / DV)			
K2: S = (D/D0)^n, D = ΛCDM linear growth	n = 3.5: −3.95 / −11.60 / −8.07	holds	n = 3.7 / 3.5 / 3.6: −6.56 / −11.60 / −10.27	+1.04 / +2.25 / +0.57
K3: S = a^n	n = 2.5: +13.0 / +12.3 / +13.7	disfavoured	unchanged	+20.6 / +26.2 / +24.5
K1: collapsed fraction (Press–Schechter; fast code only)	10^{13} M_⊙/h: +119 / +22 / +133	excluded	+39 / +2.8 / +30	—

- **K2 at its minima:** δ0 = 0.769–0.795, h = 0.666–0.677, Ω_m = 0.310–0.319; n within 3.40–3.76 at $\Delta\chi^2 = 1$. The full check reproduces K2 and K3 within 0.004 in χ^2.
- **Effective history (K2):** w_eff = −0.83/−0.78/−0.80 today, crossing −1 at z = 0.32/0.49/0.40, about −1.2 at z = 3.

7.4 Growth with the read history (T8c, K2 at n = 3.5)

- **Growth following H_read** (CAMB-PPF, Planck's A_s and n_s): compatible with S1, S2 and S3; σ8 = 0.793–0.794, S8 = 0.815–0.819, read age 13.75 Gyr.
- **Growth of the bare universe:** incompatible even with a free amplitude (S2: χ^2 36.0 for 7 points); bare age 17.3 Gyr.
- **Equivalence:** with the same ingredients and this perturbation rule, the growth coincides with that of a history with the same H(z). Expansion and growth data alone do not tell the clock reading from a substance.

7.5 Tracer built from measured growth, and a check (T8d, T8e, T8f)

- **What was done:** instead of the ΛCDM growth curve, a curve fitted to measured fσ8 (S1 or S2), with growth not faster than in the matter era ($f \leq 1$, an assumption of the tested family). Expansion and growth were then fitted together, in the approximation that they are independent (the joint covariance of DESI growth and DESI BAO is not treated).
- **Result (T8e):** better than ΛCDM plus a free growth curve by 8.7 / 12.7 / 10.9 with DESI growth (S2) and 8.2 / 11.4 / 9.8 with SDSS (S1), with as many parameters as CPL plus the curve. These results are preliminary. σ8 today sits at the bound set by $f \leq 1$.
- **Check (T8f):** after rescaling the CAMB prediction to the fitted σ8(0), the growth shape passes both registered checks at the tested redshifts for DESI growth with Pantheon+ and Dovekie. Union3 slightly exceeds the pointwise limit: 1.07 for a limit of 1. Separately, the fitted present-day amplitude differs by less than 0.75 % from CAMB's prediction with the adopted Planck A_s (not a full statistical test). With SDSS growth the check does not pass.
- **Details,** including T8d and the sensitivity to the growth curve: NOTA_ESITO_T8.md.

7.6 Self-consistent version: the clock follows its own growth (T8g; NOTA_ESITO_T8g.md)

- **Definition:** S = (D/D0)^n, where D is the linear growth computed in the read history itself. The fixed point is found by iteration, with no growth curve fitted and no growth data in the fit. Parameters: h, ω_b, Ω_m and n.
- **Expansion (C3):**

	Pantheon+	**Union3**	**Dovekie**
$\Delta\chi^2$ vs ΛCDM	−7.03	−11.85	−10.59
$\Delta\chi^2$ vs CPL	+0.57	+2.00	+0.25
n	3.707	3.575	3.692
full check − fast code, evaluated at the fast-code minimum	+0.0042	+0.0045	+0.0040

Zone: holds. These are the best points found from three starts; neither uniqueness of the fixed point nor a global minimum is demonstrated.

- **Growth as a test** (CAMB-PPF in the read history, central values of Planck's A_s and n_s, no free amplitude):
- σ8 = 0.797–0.808 and S8 = 0.820–0.824;

- χ^2 6.26–6.80 for SDSS (ΛCDM 6.6), 3.77–4.10 for DESI (4.0) and 0.03–0.05 for 6dFGS (0.02), compatible in all sets;
- a free amplitude would be within 1.51σ of 1.
- The growth comparison is still partly correlated with the expansion data.
- **Internal coherence:** the normalized growth D(z)/D(0) of the fixed point agrees with CAMB's cold-matter-plus-baryon growth within $2\cdot10^{-4}$ at seven checked redshifts between 0 and 2, at k = 0.05 Mpc^{-1}. This does not check other scales, the absolute amplitude or f. The CAMB curve is not itself reinserted into the clock law.
- **Effective history:** w_eff = −0.83/−0.80/−0.83 today and −1.04/−1.00/−1.04 at z = 0.5, so it crosses −1 between z = 0 and 0.5.
- **Limits:** the clock law, the distance rule and the linear, scale-independent growth are assumptions, and there is no clock-specific prediction yet.

8. Growth of structure for the MM 2.32 family (G0 and part B; NOTA_ESITO_G0.md, NOTA_ESITO_B.md)

- **G0, selected histories from T1–T7 with Planck's amplitude:**
- all are compatible with fσ8 from SDSS, DESI DR1 and 6dFGS under the registered rule, except one case in tension at 2.1σ (thawing with Union3, full SDSS likelihood);
- the measurements do not distinguish them from ΛCDM ($|\Delta\chi^2| \le 0.96$).
- **B1, the coupled MM Boltzmann code:** at three chain points, its growth normalized today agrees with CAMB on the same background within the registered thresholds. The background agrees within 2.2×10^{-6} and the MM/ΛCDM growth ratio within 6.7×10^{-4} on the tested grid. This comparison does not isolate dark-sector perturbations or verify the σ8 normalization used in B3.
- **B3, 400 samples of the published 2.32 chain:**
- median σ8 = 0.790 (95 %: 0.771–0.800), below ΛCDM with the same parameters in every sample (median −0.86 %);
- no paired $\Delta\chi^2$ exceeds 4;
- 2 % of the samples are in tension in SDSS growth, only through the amplitude (at most 2.44σ).
- **B2, noise of the readings:** at the tested points, the rms response of matter to record-count noise is bounded, per Fourier mode, by 1.1×10^{-5} rms(δ_m) for density and by 3.8×10^{-5} f rms(δ_m) for $\delta' = d\delta/d\ln a$. These bounds use the adopted source model and normalizations, in the branch with no additional vacuum component and in linear sub-horizon theory without back-reaction or relativistic terms. In this approximation the noise is negligible (registered threshold 10^{-3}). It is not a bound on fσ8.
- **Scope:**
- selected histories, with no MM-specific perturbations in G0;
- A_s fixed to Planck's central value;
- the growth data are not all independent of the expansion data.
- Full list of limits, and the two earlier numerical versions of B2 that are not used: NOTA_ESITO_B.md.

9. How MM 2.32 should be read now (blocchi/09_lettura_della_2.32)

- **The translation, not the premises:** 2.32 translated record counts into an energy density and a pressure (declared there as an imported effective postulate). The constraint w ≥ −1 comes from that translation, not from the author's premises.
- **The published version stands as computed:** its evaluation is the one in section 4 ("in tension").
- **Still to do:** a translation of the 2.32 counts into the clock reading of section 7 has not been written yet. Renaming the same H(z) as a "rate" would not change its w.

10. What the hypothesis answers, and what is open

It answers together, as coherence and not as independent confirmations:

1. late acceleration without Λ, with good fits to DESI DR2 BAO, CMB distance priors and three supernova samples;
2. an effective w ≈ −0.8 today crossing −1 at z ≈ 0–0.6, in the range favoured by these parametrizations and data;
3. a late onset, because the tracer follows structure growth; how late depends on the chosen form and the fitted n, and structures also grow earlier. The parametric reconstructions without MM locate the best-constrained deviations at z ≈ 0.4–1 on the same data: a qualitative coherence;
4. a self-consistent version whose own growth, with Planck's amplitude, is compatible with the measurements;
5. a background and early dark fraction under control (not the full early-universe physics or the CMB spectrum);
6. read as one possible interpretation, two working premises of the author: proper time tied to local structure, and space changing only through time. The code uses a global cosmological curve, not local structure or

environment.

The clock is tied to a quantity that exists and is accessible through growth observations (here not statistically independent of all the expansion data); a substance with the same H(z) would have to be postulated for the purpose. This is the author's motivation for studying the hypothesis, not a result of the χ^2 values. The observed growth does not by itself support the particular law linking it to the clock.

Open:

- the exponent n is fitted: 3.5–3.7 for K2, 3.575–3.707 for T8g, up to 3.93 for T8e with SDSS;
- the law linking growth and clock, and the distance rule, are assumptions;
- the joint covariance of DESI growth and DESI BAO is not treated;
- growth enters the clock law as one scale-independent linear factor;
- there is no clock-specific prediction yet, for example a dependence on environment;
- the translation of the 2.32 counts into the clock reading is not yet written.

11. Reproduction

- **blocchi/** holds each block with its scripts, results, registrations, notes, REGISTRO.txt and the checker's prompts and reports. Working files and folder names are in Italian.
- **dipendenze/mm_core** holds the Python modules of the MM 2.32 fitting core used here (for example distance_priors.py with the CHW means, symmetrized errors and correlation matrix). The native HyRec and coupled CLASS components are in the MM 2.32 kit.
- **COLLAUDO.md** reports a re-evaluation of the main numbers from the stored minima on the author's machine, with the working-folder code, byte-identical to the code in blocchi/. The scripts contain absolute paths of that machine, to be adapted.
- **Software:** Python 3.13, NumPy, SciPy, CAMB 2.0.4, Astropy (for Union3 FITS).
- **IMPRONTE.sha256** lists the SHA-256 of every file in the deposit except itself.

12. Records

- **In each block:**
- REGISTRAZIONE_* (registrations), CORREZIONI_* (corrections after each check round), AGGIUNTA_* (additions);
- BOZZA_* (drafts), NOTA_ESITO_* (final notes) and REGISTRO.txt (time-stamped registry);
- astra_*/PROMPT_*.md and RAPPORTO.md (the checker's prompts and reports).
- **The blocks:**
- 01 T1 (nine check rounds);
- 02–05 T2–T5;
- 06 T6–T7 (premise withdrawn);
- 07 R1;
- 08 T8 (T8a–T8g);
- 09 reading of MM 2.32;
- 10 G0;
- 11 part B.
- **The checks of this text** are in astra_deposito_*.

Glossary

- **Bare universe:** matter and radiation (with neutrinos), no dark energy; H_bare = 100 √G.
- **Read rate:** the expansion rate as read by the observer's clock, H_read = H_bare (1 + δ0 S).
- **Tracer S(z):** a function of structure growth, 1 today and 0 without structure.
- **MCD (in the author's language):** a local structure of the relational network; here represented only through the global growth of structure.
- **CPL (w0wa):** the dark-energy equation of state w(a) = w0 + wa (1 − a).
- **$\Delta\chi^2$:** difference in χ^2 from ΛCDM at the best points; negative means a better fit.
- **fσ8:** growth rate times the amplitude of matter fluctuations on 8 h^{-1} Mpc. **S8** = σ8 (Ω_m/0.3)^0.5.
- **b:** a free amplitude multiplying the predicted fσ8, used to test the shape of growth separately.
- **PPF:** parametrized post-Friedmann treatment of the dark sector, which lets w cross −1 in CAMB.

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